Analog Electronics Lab

Complete Viva Preparation Guide (Detailed)

Contains detailed notes for all 12 experiments.

Contents

1 Experiment 1: Half Wave Rectifier (With & Without Filter) 2

2 Experiment 2: Full Wave Rectifier (With & Without Filter) 3

3 Experiment 3: Transistor Amplifier with Self-Bias 4

4 Experiment 4: Transistor Amplifier with Potential Divider Bias 5

5 Experiment 5: Two-Stage R-C Coupled Amplifier 6

6 Experiment 6: Inverting and Non-Inverting Amplifier (Op-Amp) 7

7 Experiment 7: Differentiator and Integrator using Op-Amp 8

8 Experiment 8: Summing Amplifier using Op-Amp 9

9 Experiment 9: Logarithmic & Anti-logarithmic Amplifier 10

10 Experiment 10: Phase Shift Oscillator using Op-Amp 11

11 Experiment 11: Multivibrators (Monostable & Astable) using IC 555 12

12 Experiment 12: Schmitt Trigger circuit using IC 555 13

1 Experiment 1: Half Wave Rectifier (With & Without Filter)

Components Used: PN Junction Diode, Step-down Transformer (e.g., 12-0-12V), Load Resistor (R_L), Filter Capacitor (C), CRO, Multimeter.

1. Purpose

To study the conversion of an Alternating Current (AC) signal into a Direct Current (DC) signal using a single PN junction diode, and to analyze the Ripple Factor and Voltage Regulation with and without a capacitor filter.

2. Detailed Working Principle

- **Without Filter:** The step-down transformer provides a lower voltage AC to the diode. A diode only conducts when forward-biased (anode is positive relative to cathode). During the positive half-cycle of the AC input, the diode acts as a closed switch, passing the waveform to the load resistor (R_L). During the negative half-cycle, it is reverse-biased (acts as an open switch) and blocks the current. The output is a series of positive pulses separated by zero-voltage gaps (Pulsating DC).
- **With Capacitor Filter:** A high-value capacitor is connected in parallel with R_L. During the positive half-cycle, the capacitor charges up to the peak input voltage (V_m). When the input voltage starts dropping and goes into the negative half-cycle, the diode becomes reverse-biased. The capacitor now acts as a temporary battery, discharging its stored energy through R_L. If the RC time constant is large enough, it discharges very slowly, maintaining a relatively high voltage until the next positive cycle recharges it. This significantly reduces the ripple.

3. Important Viva Questions & Answers

- **Q: What is the theoretical efficiency of a Half Wave Rectifier?**
  A: The maximum theoretical efficiency is 40.6%. This means only 40.6% of the input AC power is converted into useful DC power; the rest is lost as heat or remains as AC ripple.
- **Q: What is Peak Inverse Voltage (PIV) and what is it for a HWR?**
  A: PIV is the maximum reverse voltage the diode experiences when it is not conducting. For a HWR, the PIV is V_m (the peak value of the secondary voltage).
- **Q: What is Ripple Factor (γ) and what are its values?**
  A: Ripple factor measures the purity of the DC output. $\gamma = V_{ac(rms)}/V_{dc}$. For a HWR without a filter, it is roughly 1.21 (or 121%), meaning the AC component is larger than the DC component.
- **Q: What is the output frequency of the HWR ripple?**
  A: The ripple frequency is exactly equal to the input supply frequency (e.g., 50 Hz).
- **Q: Why is the HWR rarely used in practical power supplies?**
  A: Because of its very high ripple factor, low efficiency (40.6%), and low Transformer Utilization Factor (TUF = 0.287), making it highly inefficient.

2 Experiment 2: Full Wave Rectifier (With & Without Filter)

Components Used: Center-tap Transformer (e.g., 9-0-9V) or 4 diodes for Bridge Rectifier, Load Resistor (R_L), Filter Capacitor, CRO.

1. Purpose

To overcome the inefficiency of the half-wave rectifier by converting both the positive and negative half-cycles of the AC input into DC, and to study its ripple and regulation parameters.

2. Detailed Working Principle

- **Without Filter (Center-Tap):** The transformer secondary is split by a center tap (ground). During the positive half-cycle, the top half of the secondary is positive, forward-biasing Diode 1 (D_1), while Diode 2 (D_2) is reverse-biased. Current flows through D_1 and down through R_L. During the negative half-cycle, the bottom half becomes positive, forward-biasing D_2, while D_1 turns off. Current flows through D_2 and *down* through R_L again. Thus, current always flows in the same direction, yielding continuous positive output pulses.
- **With Capacitor Filter:** The capacitor charges to the peak voltage (V_m) twice as often as in the HWR. Because the ”gaps” between pulses are eliminated, the capacitor has much less time to discharge between charging cycles. This results in a significantly flatter DC output with much less ripple.

3. Important Viva Questions & Answers

- **Q: Compare the efficiency and ripple factor of FWR with HWR.**
  A: FWR is twice as efficient (maximum 81.2%) and has a much lower ripple factor (0.48 or 48% without filter) compared to the HWR.
- **Q: What is the ripple frequency of a Full Wave Rectifier?**
  A: It is twice the input frequency ($2 \times f_{in}$). For a 50Hz supply, the ripple frequency is 100Hz.
- **Q: What is the PIV for a Center-Tap FWR vs a Bridge FWR?**
  A: In a Center-Tap FWR, the PIV is $2V_m$ (twice the peak secondary voltage). In a Bridge Rectifier, the PIV is only V_m, which is why bridge rectifiers are more popular.
- **Q: What is Voltage Regulation?**
  A: It is the ability of a power supply to maintain a constant output voltage regardless of changes in load current. Formula: $\%Reg = ((V_{no-load} - V_{full-load})/V_{full-load}) \times 100$.
- **Q: Why might we prefer a Bridge Rectifier over a Center-Tap one?**
  A: A center-tapped transformer is bulky and expensive. The bridge uses a standard transformer and requires diodes with a lower PIV rating.

3 Experiment 3: Transistor Amplifier with Self-Bias

Components Used: NPN Transistor (e.g., BC547), Resistors (R_C, R_B, R_E), DC Power Supply, Multimeter.

1. Purpose

To design a self-biasing (emitter-biasing) circuit for a BJT and study its effectiveness in stabilizing the Quiescent (Q) point against variations in temperature and transistor parameters.

2. Detailed Working Principle

In an amplifier, the transistor must remain in the "active region." The DC voltage and currents (V_{CE} and I_C) that define this state form the Q-point.

- **Thermal Runaway:** As a transistor operates, it heats up. Heat increases the reverse saturation current (I_{CO}), which increases I_C. Increased I_C causes more heat, leading to a vicious cycle that can destroy the transistor.
- **Self-Bias Mechanism:** We add a resistor to the emitter (R_E). If temperature causes I_C to rise, the voltage drop across R_E ($V_E = I_E \times R_E \approx I_C \times R_E$) also rises. Since the base voltage is somewhat fixed, an increase in V_E decreases the base-emitter voltage ($V_{BE} = V_B - V_E$). A lower V_{BE} reduces the base current (I_B), which consequently forces I_C back down. This negative feedback stabilizes the Q-point.

3. Important Viva Questions & Answers

- **Q: What is a DC Load Line?**
  A: It is a straight line drawn on the transistor's output characteristics graph representing all possible combinations of I_C and V_{CE} for a given circuit. The Q-point lies on this line.
- **Q: What is the Stability Factor (S)?**
  A: It measures how sensitive the collector current (I_C) is to changes in reverse saturation current (I_{CO}), V_{BE}, or β. A lower stability factor means a highly stable circuit. Ideal $S = 1$.
- **Q: Why is it called "Self-Bias"?**
  A: Because the circuit automatically adjusts its own bias condition. The output current (I_E) creates a voltage that feeds back to the input to correct any drift.
- **Q: What is the disadvantage of the emitter resistor R_E in AC operations?**
  A: While R_E provides DC stability, it also provides negative feedback to the AC signal, significantly reducing the voltage gain. (This is fixed by using a bypass capacitor in AC amplifiers).

4 Experiment 4: Transistor Amplifier with Potential Divider Bias

Components Used: NPN Transistor, Resistors (R_1, R_2, R_C, R_E), Emitter Bypass Capacitor (C_E), Input/Output Coupling Capacitors (C_{in}, C_{out}), DC Supply, Function Generator, CRO.

1. Purpose

To construct and study a Common Emitter (CE) amplifier using Potential Divider bias, verifying its superior Q-point stability, output voltage, and calculating its AC voltage gain.

2. Detailed Working Principle

- **The Bias Circuit:** R_1 and R_2 form a voltage divider across V_{CC}. They provide a firm, constant voltage to the transistor's base. The Thevenin equivalent voltage at the base is $V_{TH} = V_{CC} \times (R_2/(R_1 + R_2))$. This design ensures that the bias is virtually independent of the transistor's inherent current gain (β), making it the most reliable biasing method for mass production.
- **AC Operation:** An AC signal is injected through C_{in}. The transistor amplifies the variations in base current, causing massive variations in collector current. This alternating collector current drops voltage across R_C, creating the amplified output signal.
- **Role of Capacitors:** C_{in} and C_{out} block DC from messing with the bias, but pass AC. C_E (bypass capacitor) provides a low-resistance path for the AC signal to ground, bypassing R_E to prevent AC gain reduction.

3. Important Viva Questions & Answers

- **Q: Why is Potential Divider bias also known as Universal Bias?**
  A: Because it provides the best stability factor ($S \approx 1$) and the operating point is largely immune to transistor replacement or temperature changes.
- **Q: What happens if the bypass capacitor (C_E) is removed?**
  A: The AC signal will flow through R_E, creating a voltage drop that opposes the input signal (negative feedback). The amplifier's voltage gain will drop drastically.
- **Q: Why does the CE amplifier introduce a 180° phase shift?**
  A: As the input base voltage goes positive, collector current increases. Since $V_{out} = V_{CC} - I_C R_C$, an increase in I_C causes a larger drop across R_C, meaning the output voltage (V_{out}) drops. Thus, input goes up, output goes down.
- **Q: What is the condition for good voltage divider action?**
  A: The base current should be very small compared to the current flowing through R_1 and R_2. Typically, we ensure that $\beta R_E \geq 10 R_2$.

5 Experiment 5: Two-Stage R-C Coupled Amplifier

Components Used: Two NPN Transistors, Biasing Resistors, Coupling Capacitors (C_c), Bypass Capacitors (C_e), Function Gen, CRO.

1. Purpose

To study the cascading of two Common Emitter amplifier stages to achieve a much higher overall voltage gain, and to plot its frequency response curve to determine the bandwidth.

2. Detailed Working Principle

A single amplifier stage might provide a gain of, say, 50. If we need a gain of 2500, we cascade two stages.

- **Coupling:** The AC output from the collector of Stage 1 is fed into the base of Stage 2. We cannot connect them directly with a wire because the high DC collector voltage of Stage 1 would destroy the base bias of Stage 2. We use an R-C coupling network (a Coupling Capacitor C_c).
- **Overall Gain:** The total voltage gain is the product of the individual gains: $A = A_1 \times A_2$. In decibels (dB), the gains add up: $A_{dB} = A_{1(dB)} + A_{2(dB)}$.
- **Frequency Response:** At very low frequencies, the capacitive reactance ($X_c = 1/2\pi fC$) of the coupling capacitors is high, acting like an open circuit and reducing gain. At very high frequencies, the parasitic internal capacitances of the transistors short the signal to ground, again reducing gain. The gain is constant only in the ”mid-band” frequencies.

3. Important Viva Questions & Answers

- **Q: What is Bandwidth?**
  A: It is the difference between the upper cut-off frequency (f_H) and lower cut-off frequency (f_L). $BW = f_H - f_L$. It is the range where the amplifier operates efficiently.
- **Q: What is the significance of the 3dB frequencies?**
  A: The -3dB cut-off frequencies are the points where the voltage gain drops to $1/\sqrt{2}$ (or 70.7%) of its maximum mid-band value. At this point, the output power is half of the maximum power.
- **Q: Why are RC coupled amplifiers widely used?**
  A: They have excellent, flat frequency response in the audio range (20Hz - 20kHz), are cheap, compact, and lightweight because they don’t use heavy coupling transformers.
- **Q: What is the Miller Effect?**
  A: It is the phenomenon where the internal base-collector capacitance of the transistor is multiplied by the amplifier’s gain, appearing as a much larger capacitance at the input, which severely restricts high-frequency performance.

6 Experiment 6: Inverting and Non-Inverting Amplifier (Op-Amp)

Components Used: Op-Amp IC 741, Resistors (R_{in}, R_f), Dual DC Power Supply (+15V, -15V), Function Generator, CRO.

1. Purpose

To design and verify the operation of closed-loop voltage amplifiers using an Operational Amplifier (Op-Amp), utilizing negative feedback to achieve stable, predictable gain.

2. Detailed Working Principle

An Op-Amp without feedback has a massive open-loop gain (around 200,000 for a 741 IC). Applying even a few millivolts to the input will push the output to its maximum supply voltage (saturation). To make a usable linear amplifier, we use *Negative Feedback* by connecting the output back to the negative (inverting) terminal via a resistor R_f.

- **Inverting Amplifier:** Input signal goes to the negative terminal through R_{in}. The positive terminal is grounded. Due to virtual ground, the inverting node is also at 0V. The current flowing through R_{in} continues entirely through R_f (since ideal Op-Amp inputs draw no current). This gives a gain equation of $A_v = -R_f/R_{in}$. The output is 180° out of phase.
- **Non-Inverting Amplifier:** Input signal goes to the positive terminal. Negative feedback is still applied to the inverting terminal. The output stays in phase (0° shift) with the input. Gain equation is $A_v = 1 + (R_f/R_{in})$.

3. Important Viva Questions & Answers

- **Q: What is the concept of "Virtual Ground"?**
  A: Because the Op-Amp has extremely high gain, the voltage difference between the inverting and non-inverting terminals is practically zero ($V_+ \approx V_-$). If the non-inverting terminal is grounded (0V), the inverting terminal behaves as if it is at ground potential, even without a physical connection to ground.
- **Q: What is the purpose of Negative Feedback?**
  A: It reduces the overall gain to a manageable, precise level set by resistors, increases bandwidth, reduces distortion, and improves input/output impedance.
- **Q: Can the gain of a non-inverting amplifier be less than 1?**
  A: No. Looking at the formula $1 + (R_f/R_{in})$, the minimum possible gain is 1 (which occurs when $R_f = 0$, forming a voltage follower).
- **Q: What is CMRR (Common Mode Rejection Ratio)?**
  A: It is the ability of the Op-Amp to reject noise or signals that appear simultaneously on both input terminals. Ideal CMRR is infinite.

7 Experiment 7: Differentiator and Integrator using Op-Amp

Components Used: Op-Amp IC 741, Resistors, Capacitors, Dual Power Supply, CRO, Function Generator.

1. Purpose

To study circuits that perform the mathematical operations of calculus (differentiation and integration) on time-varying input signals, commonly used in wave-shaping and analog computers.

2. Detailed Working Principle

These circuits rely on the fact that current through a capacitor is proportional to the rate of change of voltage across it ($I_c = C \cdot dV/dt$).

- **Differentiator:** A capacitor (C) is placed at the input, and a resistor (R_f) in the feedback loop. The input voltage causes a current $C(dV_{in}/dt)$ to flow. This current passes through R_f, creating the output voltage: $V_{out} = -R_f C(dV_{in}/dt)$. It outputs a scaled version of the input's derivative. (e.g., converts a triangular wave to a square wave).
- **Integrator:** A resistor (R_{in}) is at the input, and a capacitor (C_f) is in the feedback loop. The input current (V_{in}/R_{in}) charges the capacitor. The voltage across the capacitor is the integral of the current, so: $V_{out} = -\frac{1}{R_{in}C_f}\int V_{in}dt$. (e.g., converts a square wave into a triangular wave).

3. Important Viva Questions & Answers

- **Q: Why is a pure differentiator practically unstable?**
  A: A pure differentiator acts like a high-pass filter. At very high frequencies, the capacitor's reactance drops to near zero, making the circuit's gain approach infinity. It amplifies high-frequency electronic noise, becoming highly unstable and oscillating.
- **Q: How do we fix the unstable differentiator?**
  A: By adding a small resistor in series with the input capacitor to limit the maximum high-frequency gain.
- **Q: Why does a practical integrator have a large resistor in parallel with the feedback capacitor?**
  A: Op-Amps have tiny DC offset errors. In a pure integrator, even a microscopic DC error voltage will continuously charge the capacitor over time, eventually driving the output into saturation (clipping at the supply rails). The parallel resistor provides a discharge path to prevent this.
- **Q: What happens if you give a sine wave to an integrator or differentiator?**
  A: Differentiating or integrating a sine wave yields a cosine wave (a sine wave phase-shifted by 90°).

8 Experiment 8: Summing Amplifier using Op-Amp

Components Used: Op-Amp IC 741, Multiple Input Resistors (R_1, R_2, R_3), Feedback Resistor (R_f), DC Supplies, Multimeter.

1. Purpose

To build and test an analog circuit capable of mathematically adding multiple independent input voltages together to produce a single scaled output voltage.

2. Detailed Working Principle

The summing amplifier is a direct variation of the inverting amplifier.

- Multiple input voltages ($V_1, V_2, V_3...$) are connected to the inverting (-) terminal, each through its own input resistor ($R_1, R_2, R_3...$).
- The non-inverting (+) terminal is grounded, forcing the inverting terminal to a "virtual ground" (0V).
- Because the node is at 0V, the current from each input source is entirely independent of the others. For example, $I_1 = V_1/R_1$, $I_2 = V_2/R_2$.
- According to Kirchhoff's Current Law (KCL), all these currents must add together and flow through the single feedback resistor (R_f).
- The output voltage becomes: $V_{out} = -[(R_f/R_1)V_1 + (R_f/R_2)V_2 + (R_f/R_3)V_3]$.

3. Important Viva Questions & Answers

- **Q: How can this circuit be used as an Averaging Amplifier?**
  A: If you make all input resistors equal ($R_1 = R_2 = R_3 = R$), and make the feedback resistor $R_f = R/n$ (where n is the number of inputs), the output becomes the exact mathematical average of the inputs: $V_{out} = -(V_1 + V_2 + V_3)/3$.
- **Q: What is a Scaling Amplifier?**
  A: It is a summing amplifier where the input resistors are of different values. This allows different input signals to be weighted differently (e.g., V_1 multiplied by 2, V_2 multiplied by 5) before being added. This is the foundation of a Digital-to-Analog Converter (DAC).
- **Q: Will the inputs interfere with each other?**
  A: No. Because they meet at the virtual ground point, there is no cross-talk or interaction between the input sources.

9 Experiment 9: Logarithmic & Anti-logarithmic Amplifier

Components Used: Op-Amp IC 741, PN Junction Diode (or matched BJT pair), Resistors, Dual Power Supply, Multimeter.

1. Purpose

To construct an amplifier whose output voltage is proportional to the natural logarithm (or the exponential/anti-log) of the input voltage, commonly used for signal compression and analog computation.

2. Detailed Working Principle

These circuits depart from linear resistors and use the inherently non-linear, exponential V-I characteristic of a semiconductor PN junction. The diode current equation is $I = I_s e^{(V/V_T)}$, where V_T is thermal voltage and I_s is reverse saturation current.

- **Log Amplifier:** A diode is placed in the feedback loop of an inverting Op-Amp. The input voltage pushes a linear current (V_{in}/R) through the diode. Because of the diode's properties, the voltage that develops across it to pass this current is logarithmic. Hence, $V_{out} = -V_T \ln(V_{in}/RI_s)$.
- **Anti-Log Amplifier:** The positions are swapped: the diode is at the input, and a resistor is in the feedback loop. The input voltage causes an exponential current to flow through the diode, which then flows through the feedback resistor, generating an exponential (anti-log) output voltage.

3. Important Viva Questions & Answers

- **Q: What are the main applications of Log/Antilog amplifiers?**
  A: 1) Analog multiplication and division (since $\log(A \times B) = \log A + \log B$). 2) Compressing signals that have a massive dynamic range (like audio or light sensor data) so they can be processed by standard ADCs.
- **Q: Why are these circuits rarely built with just a simple diode in practice?**
  A: The reverse saturation current (I_s) and thermal voltage (V_T) are highly sensitive to temperature. A basic diode circuit will drift wildly if the room temperature changes. Practical circuits use matched bipolar transistors and temperature-compensating thermistors (PTC).
- **Q: Why is a Transistor (BJT) preferred over a diode in the feedback loop?**
  A: A diode only provides a logarithmic response over about 2 decades of input voltage. A transistor with its base grounded provides a very precise logarithmic relationship over a much wider range (up to 5 or 6 decades).

10 Experiment 10: Phase Shift Oscillator using Op-Amp

Components Used: Op-Amp IC 741, Three Identical Capacitors, Multiple Resistors, CRO.

1. Purpose

To generate a continuous sinusoidal AC signal at a specific frequency from a DC supply, without requiring any external input signal.

2. Detailed Working Principle

An oscillator is essentially an amplifier with positive feedback. For sustained oscillations, it must satisfy the **Barkhausen Criteria**: 1. The total phase shift around the closed loop must be 360° (or 0°). 2. The magnitude of the loop gain ($A\beta$) must be exactly 1 (to start, it should be slightly > 1).

- The Op-Amp is configured as an Inverting Amplifier, which immediately provides a 180° phase shift.
- To get the remaining 180° for positive feedback, the output is fed back to the input through a feedback network consisting of three cascaded RC high-pass filters.
- Each RC section provides a phase shift that depends on the frequency. At one specific frequency (f_o), each section provides exactly 60° of phase shift ($60^\circ \times 3 = 180^\circ$).
- Total shift = 180° (Op-amp) + 180° (RC network) = 360°. The circuit oscillates at $f_o = 1/(2\pi RC\sqrt{6})$.

3. Important Viva Questions & Answers

- **Q: Why are exactly three RC sections used?**
  A: A single RC high-pass filter can theoretically provide a maximum of 90° phase shift, but practically much less. Two sections would require nearly 90° each, which demands infinite resistance. Three sections require a manageable 60° each.
- **Q: What is the required gain of the inverting amplifier for the circuit to oscillate?**
  A: The attenuation of the three-stage RC network at the oscillating frequency is $1/29$. Therefore, the Op-Amp must have a voltage gain of at least 29 ($R_f/R_{in} \geq 29$) to make the loop gain ≥ 1.
- **Q: Where does the initial input signal come from to start the oscillations?**
  A: When power is turned on, random thermal noise is generated in the resistors. This noise contains all frequencies. The circuit selectively amplifies only the frequency that satisfies the 360° phase shift, which quickly grows into a sine wave.

11 Experiment 11: Multivibrators (Monostable & Astable) using IC 555

Components Used: Timer IC 555, Resistors (R_A, R_B), Capacitors, CRO, Trigger switch.

1. Purpose

To study the operation of the 555 Timer IC to generate precision time delays (Monostable mode) and continuous square wave pulses (Astable mode).

2. Detailed Working Principle

Inside the 555 timer are three $5k\Omega$ resistors (hence the name "555"), forming a voltage divider that sets reference voltages at $1/3V_{CC}$ and $2/3V_{CC}$ for two internal comparators. These comparators control a flip-flop, which controls the output and a discharge transistor.

- **Monostable (One-Shot):** The circuit rests in a stable LOW state. When a negative-going pulse drops below $1/3V_{CC}$ on the Trigger pin (Pin 2), the flip-flop sets, pushing the output HIGH and allowing the external capacitor to charge through R. When the capacitor voltage reaches $2/3V_{CC}$, the upper comparator resets the flip-flop. The output goes LOW, and the capacitor rapidly discharges. The time the output stays HIGH is $T = 1.1RC$.
- **Astable (Free-Running):** The Trigger (Pin 2) and Threshold (Pin 6) are tied together across a capacitor. The capacitor charges through $R_A + R_B$ until it hits $2/3V_{CC}$ (Output is HIGH). The flip-flop resets, and the capacitor discharges through R_B only, until it hits $1/3V_{CC}$ (Output is LOW). The cycle repeats infinitely.

3. Important Viva Questions & Answers

- **Q: What is the Duty Cycle in an Astable multivibrator?**
  A: It is the ratio of time the output is HIGH (T_{high}) to the total time period (T). Formula: $\%D = (R_A + R_B)/(R_A + 2R_B) \times 100$.
- **Q: Why can't a standard 555 Astable circuit achieve exactly a 50% duty cycle?**
  A: Because the capacitor charges through two resistors ($R_A + R_B$) but discharges through only one (R_B). Thus, the charging time (HIGH state) is always longer than the discharging time (LOW state). (It can be fixed by adding a bypass diode across R_B).
- **Q: What is the purpose of Pin 5 (Control Voltage)?**
  A: It allows direct access to the $2/3V_{CC}$ internal reference. Usually, we connect a small $0.01\mu F$ capacitor from Pin 5 to ground to bypass noise. By applying an external voltage here, you can modulate the output frequency.

12 Experiment 12: Schmitt Trigger circuit using IC 555

Components Used: Timer IC 555, Resistors, Function Generator (to provide AC sine wave), CRO.

1. Purpose

To use the 555 Timer IC to design a Schmitt Trigger, a circuit used to convert an irregular, noisy, or slowly varying analog signal into a clean, sharp digital square wave.

2. Detailed Working Principle

A standard comparator switches state exactly when the input crosses a single reference voltage. If the input has electrical noise, it will cross that single reference line multiple times rapidly, causing the output to jitter and produce multiple false pulses. The Schmitt Trigger solves this using **Hysteresis** (positive feedback). It features two distinct trigger levels:

- **Upper Threshold Point (UTP):** Internally set to $2/3V_{CC}$ by the 555 timer.
- **Lower Threshold Point (LTP):** Internally set to $1/3V_{CC}$.

By tying the Trigger (Pin 2) and Threshold (Pin 6) together and applying the input signal, the circuit operates as follows: If the input is rising, the output will only flip when it crosses the high $2/3V_{CC}$ mark. If noise pushes the signal slightly down, nothing happens. The output will only flip back when the signal drops all the way below the $1/3V_{CC}$ mark. The ”gap” between UTP and LTP acts as a dead zone that completely ignores noise.

3. Important Viva Questions & Answers

- **Q: What is Hysteresis voltage?**
  A: Hysteresis (V_H) is the difference between the Upper Threshold and Lower Threshold voltages. In a 555 timer, $V_H = UTP - LTP = 2/3V_{CC} - 1/3V_{CC} = 1/3V_{CC}$.
- **Q: What happens if the peak-to-peak voltage of the input signal is less than the hysteresis voltage ($1/3V_{CC}$)?**
  A: The Schmitt trigger will fail to operate. The input signal must swing high enough to cross the UTP and low enough to cross the LTP to force the output to change states.
- **Q: What are the main applications of a Schmitt Trigger?**
  A: 1) Squaring up a sine wave or triangular wave into a perfect digital square wave. 2) Switch debouncing (ignoring the mechanical bounce when a physical button is pressed). 3) Noise removal in digital communications.